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How Time-Delayed IkB Feedback Controls NFkB Oscillation Behavior

Introduction

Nuclear factor kappa-B (NFkB) is a transcription factor that helps control inflammatory and immune responses. Under rest, NFkB is held inactive in the cytoplasm by kappa-B (IkB) proteins. When cells receive inflammatory signals, IkB kinase (IKK) enzymes are activated and target IkB to degrade them. This allows the NFkB protein to move into the nuclei and turn on downstream genes that help the cells adapt to the signals. Since prolonged NFkB activity is connected with chronic inflammation and tissue damage, cells have regulatory mechanisms that control how long NFkB stays active. The regulatory strategy used is negative feedback. In this case, the activation of NFkB induces the transcription of new IkB proteins that inactivate NFkB which eventually decrease NFkB levels. This lets the system return to its resting state.

The question we addressed for this project is how negative feedback timing and feedback strengths control NFkB activity during inflammation. Different IkB proteins can regulate NFkB on different timescales and feedback strengths which lead to faster or slower returns to baseline. To explore this question, we developed a model of NFkB regulation that includes two IkB feedback systems. Our goal is to investigate how delayed negative feedback and dual feedback strengths interact to shape NFkB dynamics during inflammation.

Results

We built a delay differential equation model of the NFkB signaling pathway of the NFkB signaling pathway that includes three main components: NFkB, kappa-B Alpha (IkBa), and kappa-B Epsilon (IkBe). Each species is represented by a coupled delay equation that tracks how its concentration changes over time. The model includes reversible complex formation between NFkB and each IkB species, constitutive degradation, IKK-mediated degradation, promoter binding and unbinding, and synthesis of new IkB proteins following promoter activation. Because all species are modeled simultaneously, changes in one component influence the others through binding, degradation, and delayed synthesis.

A single promoter delay (tau) is used to represent the time between NFkB promoter activation and newly synthesized IkB proteins. This delay is used numerically by storing the solution history over a window of length tau and using promoter-bound states when computing current synthesis rates. IkBa and IkBe use the same delay tau; as a result, transcription takes about the same time for both feedback loops. That being said, the two loops differ not by their delay times but by having different synthesis and degradation rates: a1 and b1 for IkBa and a2 and b2 for IkBe. Since IkBa has smaller synthesis and degradation parameters, it provides faster and stronger short-term feedback while IkBe provides slower and weak long-term feedback.

All reactions occur simultaneously rather than in discrete bursts. The delay term determines how long NFkB is active before negative feedback builds up and reduces its concentration. Without this delay, feedback would occur right away and the oscillations would be strongly weakened. Simulating the coupled system over many windows lets the oscillations appear naturally from delayed synthesis, binding interactions, and negative feedback rather than being forced externally. This lets us analyze how NFkB behaves under different delay lengths and how dual feedback strengths determine oscillation amplitude and frequency.

We first simulated the NFkB network under the condition where IKK was turned on throughout the entire time period (Figure 2). All nine species exhibited an oscillatory relationship together throughout the graph (Figure 2A). These species include free NFkB, free IkBa, free IkBe, the NFkB and IkBa complex, the NFkB and IkBe complex, the IkBa promoter state, the IkBe promoter state, NFkB bound to the IkBa promoter, and NfKb bound to the IkBe promoter. Across the 500-minute window, NFkB consistently generated oscillation cycles with roughly even spacing between its peaks. The oscillation frequency was slow with cycles appearing about every 60-80 minutes. IkBa and IkBe followed a similar pattern with peaks and troughs that aligned with NFkB peaks and baselines. This shows that these three species oscillated together at predictable intervals. Since Figure 2A includes all of the species, it is visually complex to look at; as a result, Figure 2B contains just NFkB, IkBa, and IkBe making the oscillatory relationships easier to observe. Free NFkB starts at a high level and rapidly drops after IKK activation. After this initial drop, NFkB concentration oscillates to peaks that are about one fifth of its initial concentration and declines throughout the time series. At the start of the time series, both IkBa and IkBe reached their highest concentrations and dropped back down once NfKB returned to its baseline. After NFkB entered its oscillatory phase, free IkBa and free IkBe peaked during NFkB's baselines and dropped during NFkB's peaks. Free IkBa and free IkBe oscillated at the same rate, but IkBe reached higher concentrations for longer periods compared to IkBa. Back to figure 2A, the NFkB and IkBa complex and NFkB and IkBe complex showed an oscillatory relationship similar to that of the NFkB and the IkB proteins. Both complexes began to rise rapidly while NFkB began to decrease rapidly from its initial spike. The complexes then started to oscillate where both complexes peaked during NFkB's baselines and dropped during NFkB's peaks. The difference between the complexes is that the NFkB and IkBa complex had a higher baseline and peak compared to free IkBa. Similarly, the NFkB and IkBe complex also had a higher baseline and peak compared to free IkBe. Both IkBa promoter states and IkBe promoter states remained at very low concentrations but still exhibited oscillatory behavior dependent on NFkB activity. After each NFkB peak, both promoter states increased very slightly then declined once NFkB returned to its baseline. The two promoter signals appeared about the same in their magnitudes. We would also like to note that only the IkBe promoter state is shown in Figure 2A since the IkBa promoter state completely overlaps it, making the two signals indistinguishable on the graph.

We next tested how changing the promoter delay affects NFkB oscillations (Figure 3). NFkB showed different time series depending on the value of tau. When tau was short (10 minutes), the oscillations had smaller amplitudes and were more frequent and remained near the baseline after its initial peak across time (Figure 3A). When tau was increased to 30 minutes, the oscillations were less frequent, had higher peak amplitudes, and longer periods across time (Figure 3B). To visualize this trend, we plotted the peak-to-peak oscillation amplitude of NFkB as a function of tau across several simulations (Figure 3C). NFkB amplitude increased steadily with larger delay times. This shows a direct relationship between tau and oscillation strength.

We also tested how changing IKK stimulus strength influenced NFkB behavior (Figure 4). When IKK amplitude was low (1.0), the oscillations were small with shallow peaks that stayed close to the baseline after the initial activation (Figure 4A). When IKK amplitude was higher (3.0), NFkB oscillations had higher peaks and were slightly more frequent compared to the oscillations with a lower IKK amplitude. To further illustrate this relationship, we plotted NFkB peak-to-peak amplitudes against varying degrees of IKK stimulus strength across several simulations (Figure 4C). NFkB amplitude increased gradually as IKK amplitude increased, which again shows a direct relationship between stimulus intensity and oscillation strength.

Discussion

Our simulations show that NFkB oscillations are directly shaped by the timing and strength of delayed IkB feedback. When promoter delay was short, NFkB returned to baseline quickly and the oscillations were weak because the negative feedback from new IkB arrived almost immediately counteracting NFkB accumulation. When the delay increased, feedback was slower meaning NFkB stayed active longer causing IkB to accumulate more slowly. This delays its rebinding to NFkB and results in stronger and slower oscillations. Comparably, higher IKK stimulus strength produced larger oscillatory peaks because NFkB stayed active longer before IkB levels accumulated enough to suppress it, whereas lower IKK amplitudes generated weaker oscillations that dampened sooner. With a smaller stimulus, NFkB activation is less intense, so IkB proteins can quickly rebind and deactivate NFkB activity. As a result, low IKK levels limit how much NFkB can accumulate between the pulses, decreasing peak amplitude and oscillation frequency. These findings answer our initial question by showing that both promoter delay and stimulus intensity act as timing controls that determine how pronounced and sustained NFkB oscillations become. Furthermore, the time dependent negative feedback from the IkB proteins is pivotal to this behavior. IkBa provides fast, short-term time control that suppresses NFkB shortly after IKK activation leading to weaker NFkB oscillatory behavior. On the other hand, IkBe provides slower, long-term time control that dampens NFkB after IkBa has been activated leading to stronger NFkB oscillatory behavior. Together, these two delayed feedback loops allow oscillation to emerge naturally from the model rather than by forcing some input. This confirms that time-delayed regulation alone can generate rhythmic NFkB behavior.

Biologically, this suggests that cells may tune NFkB signaling by regulating how quickly IkBa and IkBe promoters respond and how strongly the pathway is simulated. Fast IkBa feedback prevents excessive early NFkB activity. Contrastingly, slower IkBe accumulation sustains longer suppression which ensures that NFkB does not stay elevated for longer periods of time. The direct relationship between oscillation strength and promoter delay shows that transcriptional timing influences how long NFkB stays active inside the nucleus. This can affect downstream gene expression during inflammation. The stimulus strength of IKK is also biologically relevant since a stronger IKK is representative of a stronger inflammatory or stress signal. These naturally produce NFkB oscillations. This means that cells can scale the intensity of their hormonal and immune response based on how severe the stimulus is. Mild activations occur during weak stress and stronger activations occur during major inflammatory events. All in all, oscillatory NFkB behavior can help cells avoid excessive immune activation while still allowing strong initial responses. That being said, NFkB oscillations are not random. They give cells a controlled way to activate immune genes strongly at first and then reduce activity later, preventing constant NFkB signaling that would lead to chronic inflammation.

Figure 1

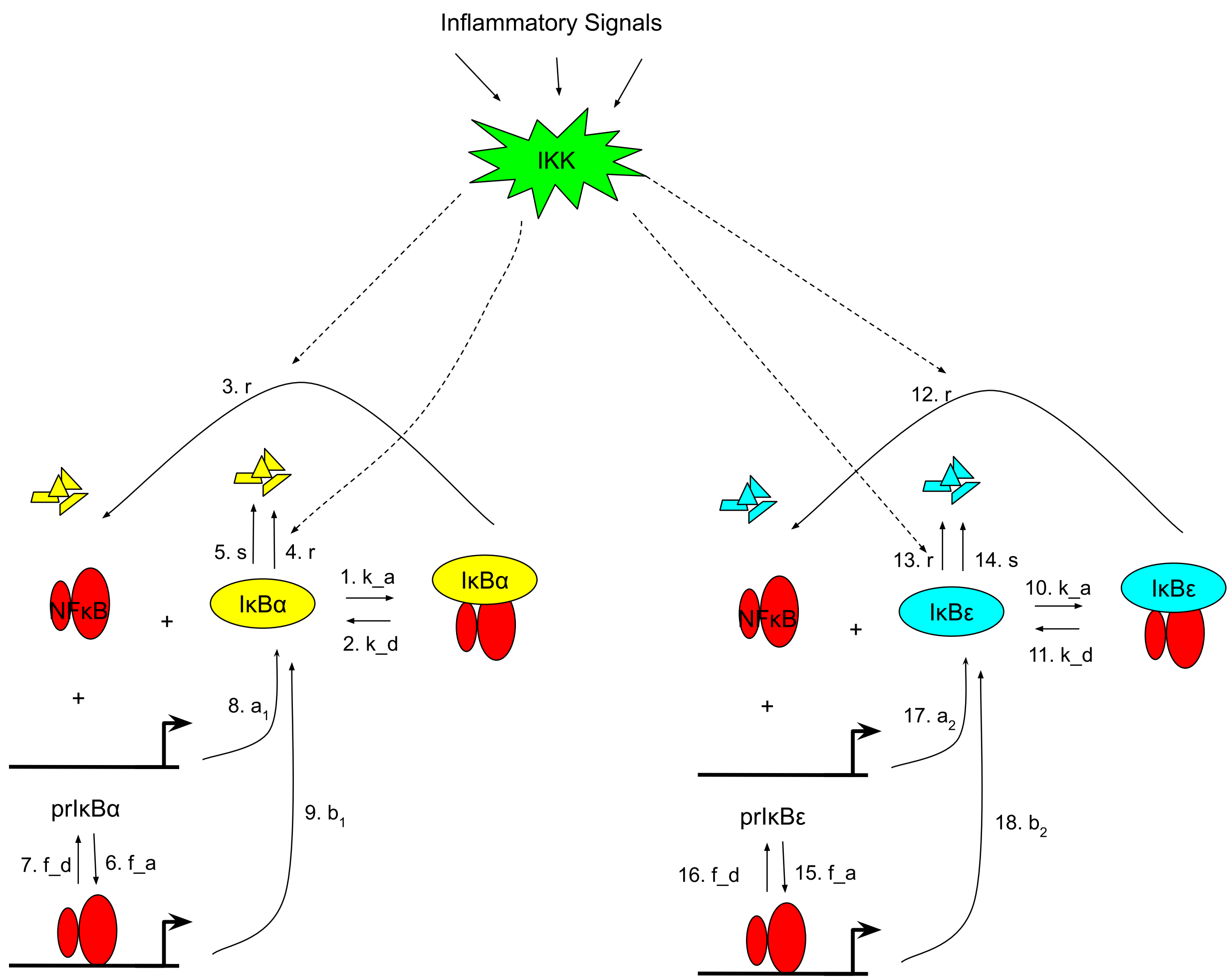

Figure 1: Network model of NFκB signaling. Inflammatory stimuli activate IKK, which promotes IkB degradation and releases NFκB from its cytoplasmic complexes. Released NFκB activates transcription of IkBa and IkBe promoters, creating two negative feedback loops. IkBa provides rapid feedback that generates sharp oscillations, while IkBe accumulates more slowly and dampens long-term NFκB activity.

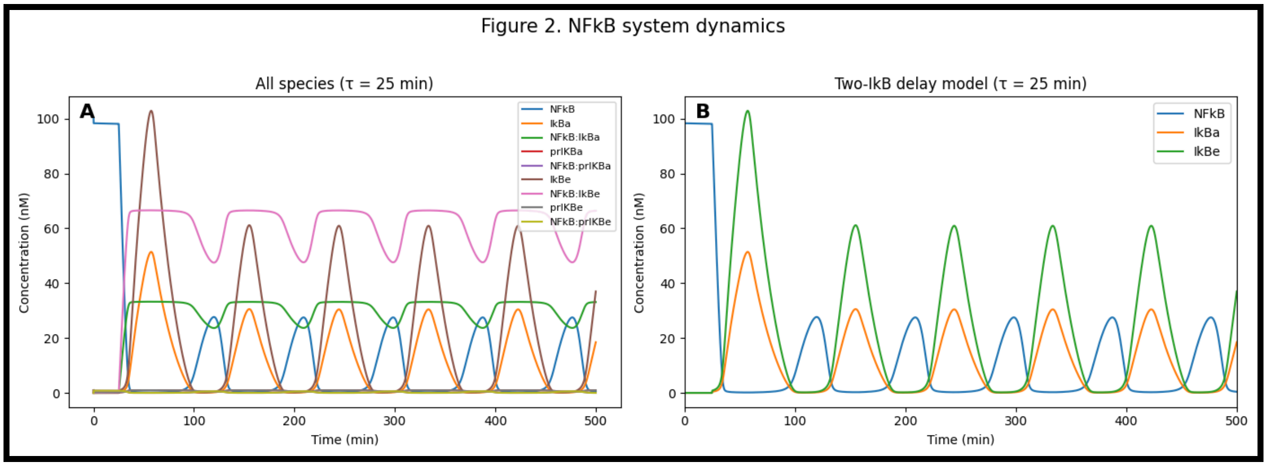

Figure 2A: Time series from the full NFκB model showing all major species, including free NFκB, IkB proteins, and promoter-bound complexes (tau = 25 min). The plot shows how each component rises and falls over time following IKK activation.
Figure 2B: Time series of NFκB, IkBa, and IkBe concentrations under sustained IKK activation (tau = 25 min). Oscillations occur because IkBa provides fast negative feedback, while IkBe provides slower feedback that shapes the long-term dynamics.

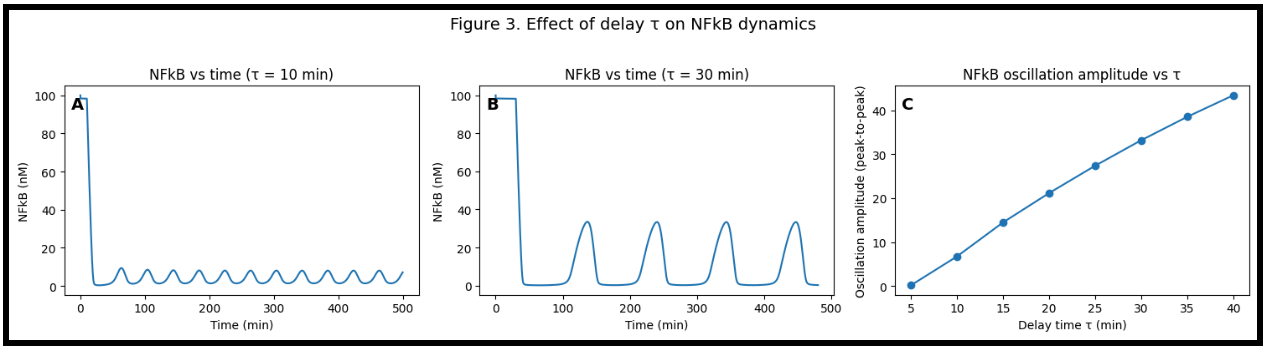

Figure 3A: Time series of NFκB concentration with a short promoter delay (τ = 10 min). Fast IkBa feedback creates small, rapid oscillations after activation.
Figure 3B: Time series of NFκB concentration with a longer promoter delay (τ = 30 min). With a longer delay, oscillations become larger and more spread out because NFκB stays active longer before feedback reduces it.
Figure 3C: Sensitivity aggregate analysis showing NFκB oscillation amplitude as a function of promoter delay τ. Each point summarizes a full time series simulation. Larger delay values lead to stronger oscillations, meaning promoter delay helps control the strength and timing of NFκB rhythmic behavior.

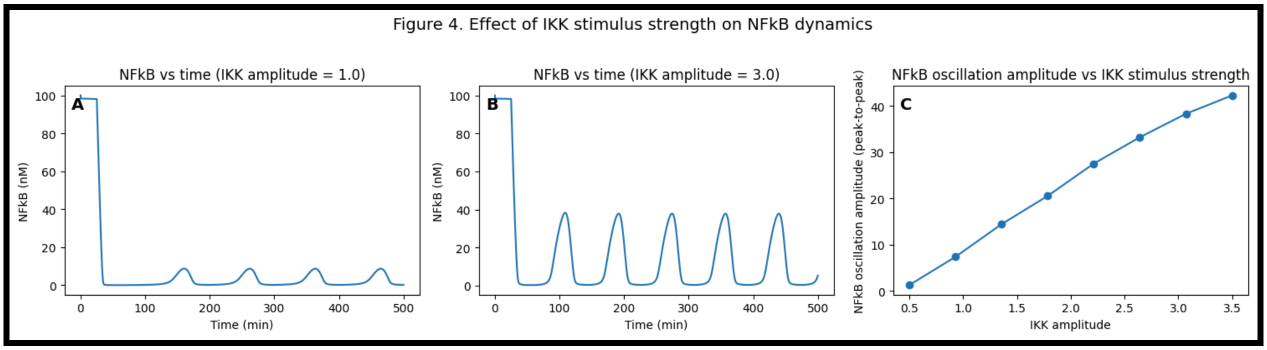

Figure 4A: Time series of NFκB concentration with a weak IKK stimulus (IKK amplitude = 1.0). Lower stimulus strength produces smaller NFκB peaks over time.
Figure 4B: Time series of NFκB concentration with a stronger IKK stimulus (IKK amplitude = 3.0). Higher stimulus strength produces larger oscillatory peaks because NFκB remains more strongly activated before feedback reduces it.
Figure 4C: Aggregate dose–response plot showing NFκB oscillation amplitude as a function of IKK amplitude. Each point summarizes a full time series simulation. Increasing IKK amplitude consistently increases oscillation strength, showing that stimulus intensity directly controls NFκB activation dynamics.

Appendix

Base Parameters

Parameters	Value	Description
k_a	0.1 minute^{-1}	Association rate of NFkB and IkB (complex formation)
k_d	0.0006 minute^{-1}	Dissociation rate of NFkB:IkB complexes
r	0.01 minute^{-1}	IKK-mediated degradation rate of IkB and NFkB:IkB complexes
g	0.012 minute^{-1}	Constitutive degradation rate of free IkB
f_a	10 minute^{-1}	Binding rate of NFkB to promoters
f_d	200 minute^{-1}	Unbinding rate of NFkB from promoters
a1	0.00185 minute^{-1}	Basal synthesis rate of IkBa from the free promoter (delayed)
b1	5.0 minute^{-1}	NFkB-induced synthesis rate of IkBa from NFkB-bound promoter (delayed)
a2	0.005 minute^{-1}	Basal synthesis rate of IkBe from the free promoter (delayed)

b2	10.0 $minute^{-1}$	NFkB-induced synthesis rate of IkBe from NFkB-bound promoter (delayed)
IKK_on_time	0 minutes	Time when IKK is turned on
IKK_off_time	1000 minutes	Time when IKK is turned off
IKK_amplitude	2.2 concentration units	Amplitude of IKK activity
tau	25 minutes	Delay between promoter binding and appearance of new IkB protein

Initial Concentrations

Parameter	Value	Description
NFKB_0	100 μM	Initial concentration of free NFkB
IKBa_0	0 μM	Initial concentration of free IkBa
NFKB_IKBa_0	0 μM	Initial concentration of NFkB:IkBa complex
prIKBa_0	1 μM	Initial amount of free IkBa promoter
NFKB_prIKBa_0	0 μM	Initial amount of NFkB bound to IkBa promoters
IKBe_0	0 μM	Initial concentration of free IkBe
NFKB_IKBe_0	0 μM	Initial concentration of NFkB:IkBe complex
prIKBe_0	1 μM	Initial amount of free IkBe promoter
NFKB_prIKBe_0	0 μM	Initial amount of NFkB bound to IkBe promoter

The history of all variables for times before zero up to time minus tau is constant at the initial values.

Chemical Reactions

$$\text{(R1)}\quad \mathrm{NF}\kappa\mathrm{B} + \mathrm{IkB}_\alpha \xrightarrow{k_a} \mathrm{NF}\kappa\mathrm{B{:}IkB}_\alpha$$

$$\text{(R2)}\quad \mathrm{NF}\kappa\mathrm{B{:}IkB}_\alpha \xrightarrow{k_d} \mathrm{NF}\kappa\mathrm{B} + \mathrm{IkB}_\alpha$$

$$\text{(R3)}\quad \mathrm{IKK} + \mathrm{NF}\kappa\mathrm{B{:}IkB}_\alpha \xrightarrow{r} \mathrm{IKK} + \mathrm{NF}\kappa\mathrm{B}$$

$$\text{(R4)}\quad \mathrm{IKK} + \mathrm{IkB}_\alpha \xrightarrow{r} \mathrm{IKK}$$

$$\text{(R5)}\quad \mathrm{IkB}_\alpha \xrightarrow{g} \varnothing$$

$$\text{(R6)}\quad \mathrm{NF}\kappa\mathrm{B} + \mathrm{prIkB}_\alpha \xrightarrow{f_a} \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\alpha$$

$$\text{(R7)}\quad \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\alpha \xrightarrow{f_d} \mathrm{NF}\kappa\mathrm{B} + \mathrm{prIkB}_\alpha$$

$$\text{(R8)}\quad \mathrm{prIkB}_\alpha \xrightarrow{a_1} \mathrm{prIkB}_\alpha + \mathrm{IkB}_\alpha$$

$$\text{(R9)}\quad \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\alpha \xrightarrow{b_1} \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\alpha + \mathrm{IkB}_\alpha$$

$$\text{(R10)}\quad \mathrm{NF}\kappa\mathrm{B} + \mathrm{IkB}_\varepsilon \xrightarrow{k_a} \mathrm{NF}\kappa\mathrm{B{:}IkB}_\varepsilon$$

$$\text{(R11)}\quad \mathrm{NF}\kappa\mathrm{B{:}IkB}_\varepsilon \xrightarrow{k_d} \mathrm{NF}\kappa\mathrm{B} + \mathrm{IkB}_\varepsilon$$

$$\text{(R12)}\quad \mathrm{IKK} + \mathrm{NF}\kappa\mathrm{B{:}IkB}_\varepsilon \xrightarrow{r} \mathrm{IKK} + \mathrm{NF}\kappa\mathrm{B}$$

$$\text{(R13)}\quad \mathrm{IKK} + \mathrm{IkB}_\varepsilon \xrightarrow{r} \mathrm{IKK}$$

$$\text{(R14)}\quad \mathrm{IkB}_\varepsilon \xrightarrow{g} \varnothing$$

$$\text{(R15)}\quad \mathrm{NF}\kappa\mathrm{B} + \mathrm{prIkB}_\varepsilon \xrightarrow{f_a} \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\varepsilon$$

$$\text{(R16)}\quad \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\varepsilon \xrightarrow{f_d} \mathrm{NF}\kappa\mathrm{B} + \mathrm{prIkB}_\varepsilon$$

$$\text{(R17)}\quad \mathrm{prIkB}_\varepsilon \xrightarrow{a_2} \mathrm{prIkB}_\varepsilon + \mathrm{IkB}_\varepsilon$$

$$\text{(R18)}\quad \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\varepsilon \xrightarrow{b_2} \mathrm{NF}\kappa\mathrm{B{:}prIkB}_\varepsilon + \mathrm{IkB}_\varepsilon$$

Ordinary Differential equations

- d(NFKB)/dt
  - - k_a * NFKB * IKBa + k_d * NFKB_IKBa + r * IKK * NFKB_IKBa
  - - k_a * NFKB * IKBe + k_d * NFKB_IKBe + r * IKK * NFKB_IKBe
  - - f_a * NFKB * prIKBa + f_d * NFKB_prIKBa
  - - f_a * NFKB * prIKBe + f_d * NFKB_prIKBe
- d(IKBa)/dt
  - - k_a * NFKB * IKBa + k_d * NFKB_IKBa + a1 * prIKBa(t - tau) + b1 * NFKB_prIKBa(t - tau) - g * IKBa - r * IKK * IKBa
- d(IKBe)/dt
  - - k_a * NFKB * IKBe + k_d * NFKB_IKBe + a2 * prIKBe(t - tau) + b2 * NFKB_prIKBe(t - tau) - g * IKBe - r * IKK * IKBe
- d(NFKB_IKBa)/dt
  - k_a * NFKB * IKBa - k_d * NFKB_IKBa - r * IKK * NFKB_IKBa
- d(NFKB_IKBe)/dt
  - k_a * NFKB * IKBe - k_d * NFKB_IKBe - r * IKK * NFKB_IKBe
- d(prIKBa)/dt
  - - f_a * NFKB * prIKBa + f_d * NFKB_prIKBa
- d(prIKBe)/dt
  - - f_a * NFKB * prIKBe + f_d * NFKB_prIKBe
- d(NFKB_prIKBa)/dt
  - f_a * NFKB * prIKBa - f_d * NFKB_prIKBa
- d(NFKB_prIKBe)/dt
  - f_a * NFKB * prIKBe - f_d * NFKB_prIKBe

Acknowledgement

Team Members

- Allison Brookhart
  - Assisted in implementing the Two-IkB model dynamics and created a Google Drawing illustrating the signaling network
- Daniel Chang
  - Contributed to coding the Two-IkB model dynamics and developed simulating explaining how promoter delay (tau) and IKK stimulus amplitude affect NFkB oscillations affect NFkB oscillation amplitudes
- Yara Castillo-Denker
  - Formulated the chemical reactions and differential equations used in the model and produced a hand-drawn network diagram